

RISTNA, Estonia

WMO: 261150

Lat: 58.9209N

Lon: 22.0664E

Elev: 8

StdP: 101.23

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.0	-10.4	-16.1	0.9	-11.9	-13.6	1.2	-9.2	12.7	2.9	11.2	2.3	2.2	90	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.5	25.3	19.8	23.4	18.9	21.7	17.9	20.6	24.0	19.5	22.5	18.5	21.1	2.4	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.1	22.5	18.3	13.2	21.3	17.4	12.4	20.3	59.4	24.1	55.7	22.6	52.5	21.1	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	9.5	8.0	DB	-15.2	27.2	4.2	2.2	-18.2	28.7	-20.7	30.0	-23.0	31.3	-26.1	32.9
			WB	-15.5	21.5	4.1	1.6	-18.4	22.7	-20.8	23.7	-23.1	24.6	-26.0	25.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-1.2	-2.0	-0.1	4.0	8.8	13.4	17.7	17.2	13.3	8.3	4.2	1.4
	DBStd	7.57	4.35	4.41	2.86	2.97	3.54	2.90	2.72	2.42	2.85	3.29	3.49	4.01
	HDD10.0	1768	348	337	312	183	64	3	0	0	7	72	175	267
	HDD18.3	4133	607	570	570	431	295	152	45	50	152	312	425	525
	CDD10.0	722	0	0	0	3	28	105	239	224	106	18	0	0
	CDD18.3	46	0	0	0	0	1	3	26	16	1	0	0	0
	CDH23.3	162	0	0	0	0	1	20	101	40	0	0	0	0
	CDH26.7	16	0	0	0	0	0	1	10	5	0	0	0	0
(14) Wind	WSAvg	3.3	3.8	3.4	3.1	2.8	2.7	2.8	2.8	3.4	3.4	3.7	3.7	4.3
(15) Precipitation	PrecAvg	618	52	40	33	31	32	50	55	73	59	66	71	56
	PrecMax	847	134	91	67	61	57	112	112	147	137	125	130	161
	PrecMin	240	13	6	3	6	5	11	1	5	5	8	5	5
	PrecStd	122	26	21	15	17	16	25	31	39	31	32	32	30
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.4	4.7	8.1	16.7	22.7	25.9	28.1	26.9	20.6	15.5	10.7	8.7
		MCWB	5.7	3.3	5.1	11.0	15.6	18.9	21.3	20.9	16.7	13.6	9.8	7.8
	2%	DB	5.4	3.9	5.2	13.2	19.7	22.1	25.9	24.1	19.1	14.4	10.0	8.0
		MCWB	4.6	3.0	3.3	8.6	13.8	16.7	20.2	19.2	16.2	12.9	9.1	7.1
	5%	DB	4.5	3.4	4.1	10.6	16.8	19.7	24.1	22.5	18.0	13.6	9.3	7.2
		MCWB	3.6	2.6	2.8	7.1	12.2	15.3	19.6	18.5	15.5	12.2	8.3	6.2
	10%	DB	3.7	2.8	3.4	8.5	14.5	17.9	22.5	21.1	17.1	12.7	8.5	6.0
		MCWB	2.9	2.1	2.3	5.9	10.9	14.7	18.7	17.8	15.0	11.4	7.5	5.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.9	3.8	5.2	11.3	16.8	19.5	22.5	21.3	17.8	14.2	10.1	8.1
		MCDB	6.2	4.2	7.5	15.8	20.6	24.8	26.2	25.5	19.4	15.0	10.4	8.4
	2%	WB	4.8	3.2	3.7	9.1	14.5	17.5	21.2	20.0	16.7	13.2	9.3	7.3
		MCDB	5.2	3.6	4.7	12.0	18.5	21.0	24.5	23.1	18.3	14.1	9.9	7.9
	5%	WB	3.7	2.7	3.1	7.5	12.8	16.1	20.2	19.1	16.0	12.4	8.5	6.2
		MCDB	4.2	3.2	3.9	10.3	16.5	18.7	23.3	21.6	17.5	13.4	9.2	6.9
	10%	WB	3.0	2.2	2.5	6.1	11.1	15.0	19.1	18.3	15.3	11.6	7.6	5.3
		MCDB	3.6	2.7	3.2	8.0	13.9	17.4	22.0	20.5	16.7	12.6	8.3	5.9
(35) Mean Daily Temperature Range	5% DB	MDBR	3.3	3.5	3.8	5.2	5.9	5.4	5.5	5.1	4.4	3.5	2.9	3.0
		MCDBR	2.7	2.7	4.5	9.0	10.0	9.3	8.4	7.3	5.4	3.3	2.5	2.4
		MCWBR	2.8	2.6	3.4	5.3	5.5	4.9	4.4	4.1	3.8	2.8	2.9	2.6
	5% WB	MCDBR	2.7	2.7	3.9	8.1	8.7	7.7	7.3	5.9	4.4	2.9	2.5	2.3
		MCWBR	2.9	2.6	3.2	5.1	5.3	4.6	4.3	3.8	3.5	2.8	2.9	2.6
(40) Clear-Sky Solar Irradiance	taub		0.291	0.320	0.339	0.374	0.368	0.364	0.393	0.385	0.355	0.339	0.322	0.281
	taud		2.205	2.270	2.328	2.334	2.428	2.477	2.419	2.424	2.491	2.408	2.231	2.136
	Ebn at Noon		515	692	799	830	864	872	834	812	783	670	465	383
	Edh at Noon		46	71	89	105	103	100	104	96	77	62	45	35
(44) All-Sky Solar Radiation	RadAvg		0.27	0.82	2.36	4.11	5.74	6.13	5.75	4.52	2.81	1.23	0.35	0.18
	RadStd		0.04	0.14	0.25	0.44	0.52	0.33	0.44	0.39	0.33	0.16	0.07	0.02

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page